

Topics Covered:

- Angular Velocity
- Introduction to Rigid Body Motion
- Lie Algebras

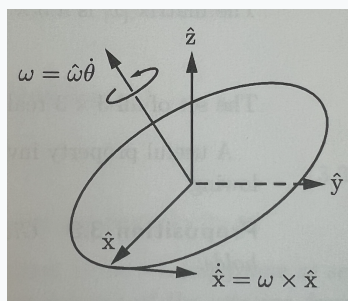
Additional Reading:

- LP 3.2.2 (Angular Velocity), 3.3.2 (Twists)
- MLS Chapter 2, Section 3.2 and Section 4

Angular Velocity

Definition: Angular Velocity

Suppose a frame with unit axes is attached to a rotating body (this frame is called the body frame). The rotation of the frame can be described as a rotation about some unit axis $\hat{\omega} \in \mathbb{R}^3$ passing through the origin of the frame.



The rate of rotation of this axis is called the **angular velocity** of the rotating body, defined by:

$$\omega = \hat{\omega} \dot{\theta},$$

where $\hat{\omega}$ is the instantaneous axis of rotation (not yet expressed in any particular reference frame) and $\dot{\theta}$ is the rate of rotation. This equation essentially describes angular velocity in terms of its direction and magnitude.

This angular velocity can be expressed in either the coordinate frame of some other fixed frame (often taken to be the world frame) or the body frame.

To solve for the *linear velocity* v at any point on a rotating object, we can use the formula:

$$v = \omega \times r$$

with r being the position vector from the axis of rotation to the point where I want to solve for velocity.

Thus, to solve for the rate of rotation for each of our coordinate axes, we have:

$$\begin{aligned}\dot{\hat{x}} &= \omega \times \hat{x} \\ \dot{\hat{y}} &= \omega \times \hat{y} \\ \dot{\hat{z}} &= \omega \times \hat{z}\end{aligned}$$

This gives us the relationship between angular velocity and the rate of change of the unit vectors of a rotating frame (which can be composed into a rotation matrix) as:

$$\dot{R} = [\omega]_{\times} R$$

with $[\omega]_{\times}$ being the skew-symmetric matrix of ω .

Definition: Skew-Symmetric Matrix

Given a vector $v = \begin{Bmatrix} a \\ b \\ c \end{Bmatrix}$ the skew-symmetric matrix is defined as:

$$[v]_{\times} = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}$$

One use of these matrices is that they transform cross products:

$$a \times b = [a]_{\times} b$$

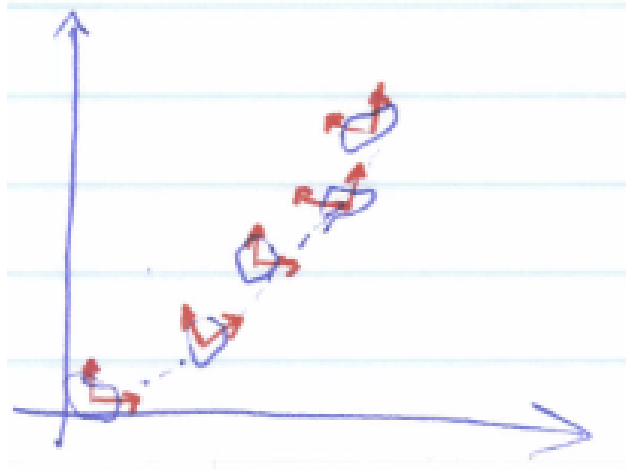
Additional properties of skew-symmetric matrices include:

1. If A is skew-symmetric, then $A^{\top} = -A$
2. If A is skew-symmetric, then $a_{ji} = -a_{ij}$

Introduction to Rigid Body Motion (Velocity)

Velocities in SE(2): So, let's start with a trajectory in SE(2), given by $g(t)$.

$$g(t) : [t_0, t_1] \rightarrow SE(2)$$



Question: What is the velocity of the body moving according to $g(t)$?

$$g(t) = \begin{bmatrix} R(\theta(t)) & d(t) \\ 0 & 1 \end{bmatrix}$$

$$\dot{g}(t) = \begin{bmatrix} \frac{d}{dt}R(\theta(t)) & \frac{d}{dt}d(t) \\ 0 & 0 \end{bmatrix}$$

where the derivative of our rotation matrix can be computed as:

$$\begin{aligned} \frac{d}{dt}R(\theta(t)) &= \frac{d}{dt} \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \\ &= \begin{bmatrix} -\sin(\theta)\dot{\theta} & -\cos(\theta)\dot{\theta} \\ \cos(\theta)\dot{\theta} & -\sin(\theta)\dot{\theta} \end{bmatrix} \\ &= \frac{\partial R(\theta)}{\partial \theta} \dot{\theta} \\ &= DR(\theta) \cdot \dot{\theta} \end{aligned}$$

Plugging this back into the velocity of our rigid body, we get:

$$\dot{g}(t) = \begin{bmatrix} DR(\theta) \cdot \dot{\theta} & \dot{d} \\ 0 & 0 \end{bmatrix}$$

To describe the change in velocity in the body frame, we can use a coordinate frame trans-

formation:

$$\begin{aligned}
 \dot{g}_B^B(t) &= g_A^B(t) \dot{g}_B^A(t) \\
 &= \left(g_B^A(t)\right)^{-1} \dot{g}_B^A(t) \\
 &= \begin{bmatrix} R^\top & -R^\top d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} DR(\theta) \cdot \dot{\theta} & \dot{d} \\ 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} R^\top DR(\theta) \dot{\theta} & R^\top \dot{d} \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

Definition: Subscript Cancellation Rule

Page 62 of Modern Robotics by Kevin Lynch and Frank Park (LP) describes the “Subscript Cancellation Rule” as follows: When multiplying two rotation matrices, if the second subscript of the first matrix matches the first subscript of the second matrix, the two subscripts “cancel” and a change of reference frame is achieved:

$$R_{ab}R_{bc} = R_{a\cancel{b}}R_{\cancel{b}c} = R_{ac}$$

A rotation matrix is just a collection of three unit vectors, so the reference frame of a vector can also be changed by a rotation matrix using a modified version of the Subscript Cancellation Rule:

$$R_{ab}p_b = R_{a\cancel{b}}p_{\cancel{b}} = p_a$$

The subscript cancellation rule also extends to transformations by considering some arbitrary reference frames a , b and c and some vector v expressed in frame b :

$$g_{ab}g_{bc} = g_{a\cancel{b}}g_{\cancel{b}c} = g_{ac}$$

and

$$g_{ab}v_b = g_{a\cancel{b}}v_{\cancel{b}} = v_a$$

where v_a is the vector v expressed in frame a .

Solving for the rotational component:

$$\begin{aligned}
 R^\top DR(\theta) \cdot \dot{\theta} &= \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}^\top \begin{bmatrix} -\sin(\theta) & -\cos(\theta) \\ \cos(\theta) & -\sin(\theta) \end{bmatrix} \cdot \dot{\theta} \\
 &= \begin{bmatrix} -\cos(\theta)\sin(\theta) + \sin(\theta)\cos(\theta) & -\cos(\theta)^2 - \sin(\theta)^2 \\ \cos(\theta)^2 + \sin(\theta)^2 & -\sin(\theta)\cos(\theta) + \cos(\theta)\sin(\theta) \end{bmatrix} \cdot \dot{\theta} \\
 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \cdot \dot{\theta} \\
 &= [\dot{\theta}]_\times
 \end{aligned}$$

This is the planar skew-symmetric matrix-version of the velocities. Thus, using this skew-

symmetric notation, we can write the body velocity as:

$$\begin{aligned}\dot{g}_B^B(t) &= \begin{bmatrix} [\dot{\theta}(t)]_{\times} & R^{\top} \dot{d} \\ 0 & 0 \end{bmatrix} && \text{(where implicitly } \dot{d}(t) = \dot{d}_B^A(t)) \\ &= \begin{bmatrix} [\dot{\theta}(t)]_{\times} & \dot{d}_B^B(t) \\ 0 & 0 \end{bmatrix} && \text{(This is called body velocity!)}\end{aligned}$$

Because $\dot{g}_B^B(t)$ depends on three variables only, it is often convenient to write it that way. Furthermore, there is another symbol for velocities of SE(2) elements; it is ξ . (the symbol xi).

The velocity ξ is written as:

$$\xi = \begin{Bmatrix} v \\ \omega \end{Bmatrix} = \begin{Bmatrix} v_x \\ v_y \\ \omega \end{Bmatrix} \quad (\xi \in \mathbb{R}^3 \text{ for SE(2)}, \xi \in \mathbb{R}^6 \text{ for SE(3)})$$

this can be thought of as the vector form of velocity for the vector form coordinates $g = \{x, y, \theta\}$.

Ok, but what is ξ in homogeneous representation?

$$\hat{\xi} = \begin{bmatrix} [\omega]_{\times} & v \\ 0 & 0 \end{bmatrix}$$

(If you want to keep the notation separate from unit vectors, we can denote this as $(\cdot)^{\wedge}$)

Thus, from vector form of a vector to homogeneous form is called “hatting”. The other way is called “unhatting” and will be denoted as $(\cdot)^{\vee}$.

The benefit of homogeneous coordinates will be that it will allow us to easily change between spacial and body velocity

$$\begin{aligned}\dot{g}_B^B &= g_A^B \cdot \dot{g}_B^A \\ &\Downarrow \\ \hat{\xi}^B &= g_A^B \cdot \hat{\xi}^A\end{aligned}$$

Lie Algebras

Notably ξ is a Lie algebra element of our Lie group $SE(2)$, i.e., $\hat{\xi} \in \mathfrak{se}(2)$. Similarly, $[\omega]_{\times} \in \mathfrak{so}(2)$. Here, the gothic font is used to denote the Lie algebra element of the Lie group. In short, the Lie algebra is the tangent space of the Lie group near the identity.

- For $SE(2)$, the Lie algebra element is denoted $\mathfrak{se}(2)$
- For $SE(3)$, the Lie algebra element is denoted $\mathfrak{se}(3)$
- For $SO(2)$, the Lie algebra is denoted $\mathfrak{so}(2)$
- For $SO(3)$, the Lie algebra is denoted $\mathfrak{so}(3)$

Definition: Lie algebra

A Lie algebra is a vector space $\mathfrak{g}$ together with an operation (called the Lie bracket) denoted $\mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ that satisfies the Jacobi identity ($x \times (y \times z) + y \times (z \times x) + z \times (x \times y) = 0$).

Notably, every Lie group gives rise to a Lie algebra, which is the tangent space at the identity.

Example:

- for $\mathfrak{so}(2)/\mathfrak{so}(3)$, the Lie algebra is the set of all possible $\dot{R}$ when $R = I$
- for $\mathfrak{se}(2)/\mathfrak{se}(3)$, the Lie algebra is the set of all possible $\dot{g}$ when $g = I$

Spatial and Body Twists

Instead of always saying “Lie algebra element”, we call this a **twist**. In particular, we will use the notation ξ_b to denote the body twist and ξ_s to denote the spatial twist, with the following definitions:

$$\begin{aligned}\hat{\xi}_s &= \dot{g}_{sb} g_{sb}^{-1} \in \mathfrak{se}(2)/\mathfrak{se}(3) \\ \hat{\xi}_b &= g_{sb}^{-1} \dot{g}_{sb} \in \mathfrak{se}(2)/\mathfrak{se}(3)\end{aligned}$$

Computing these gives us the individual formulas for body and spatial angular and linear velocity:

body twist:

$$\begin{aligned}\hat{\xi}_b &= g_{sb}^{-1} \dot{g}_{sb} = g_{b\sharp} \dot{g}_{\sharp b} = \begin{bmatrix} R^\top & -R^\top d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{R} & \dot{d} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} R^\top \dot{R} & R^\top \dot{d} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} [\omega_b]_\times & v_b \\ 0 & 0 \end{bmatrix} \in \mathfrak{se}(2)/\mathfrak{se}(3)\end{aligned}$$

spatial twist:

$$\begin{aligned}\hat{\xi}_s &= \dot{g}_{sb} g_{sb}^{-1} = \dot{g}_{s\sharp} g_{\sharp s} = \begin{bmatrix} \dot{R} & \dot{d} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^\top & -R^\top d \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \dot{R} R^\top & -\dot{R} R^\top d + \dot{d} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} [\omega_s]_\times & v_s \\ 0 & 0 \end{bmatrix} \in \mathfrak{se}(2)/\mathfrak{se}(3)\end{aligned}$$

Note that in both cases, we can get the vector form by “unhatting”:

$$\xi_b = (\hat{\xi}_b)^\vee = \begin{Bmatrix} v_b \\ \omega_b \end{Bmatrix}, \quad \xi_s = (\hat{\xi}_s)^\vee = \begin{Bmatrix} v_s \\ \omega_s \end{Bmatrix}$$

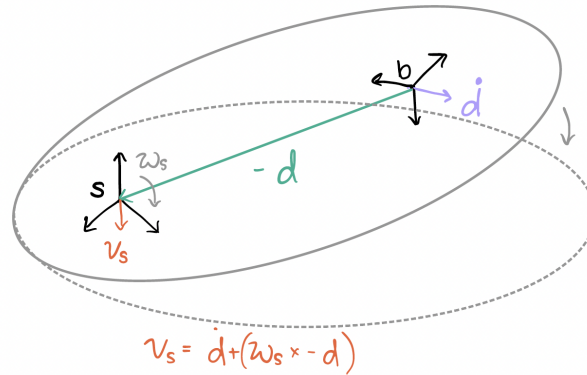
Also, note that in this formula, $v_s = -\dot{R} R^\top d + \dot{d}$ should NOT be interpreted as the linear velocity of the base-frame origin expressed in the fixed frame (that would simply be $\dot{d}$). Instead, the physical meaning is the instantaneous velocity of the body, currently at the fixed-frame origin, expressed in the fixed frame:

- ω_b : the angular velocity expressed in $\{b\}$
- ω_s : the angular velocity expressed in $\{s\}$
- v_b : the linear velocity of a point at the origin of $\{b\}$ expressed in $\{b\}$
- v_s : the linear velocity of a point at the origin of $\{s\}$ expressed in $\{s\}$

The intuition behind the expression for v_s can be derived as:

$$\begin{aligned}v_s &= \dot{d}_{sb} - \dot{R} R^\top d \\ &= \dot{d}_{sb} - [\omega_s]_\times d \\ &= \dot{d}_{sb} - \omega_s \times d \\ &= \dot{d}_{sb} + \omega_s \times -d\end{aligned}$$

This is summarized by the diagram:



Lastly, we can change our reference frame from body to spatial by using adjoint transformation:

$$\xi_s = \text{Ad}_g \xi_b \quad (\text{Note here that } g \text{ is } g_{sb})$$

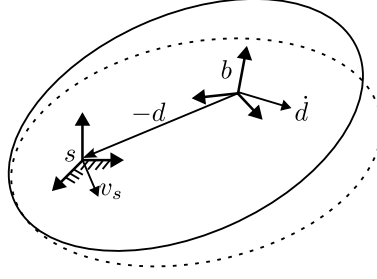
Where this adjoint is performed the same way as we learned earlier in the course:

$$\begin{aligned}
 \hat{\xi}_s &= g_{sb} \hat{\xi}_b g_{sb}^{-1} \\
 &= \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} [\omega_b]_{\times} & v_b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\top} & -R^{\top} d \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R^{\top} \dot{R} & R^{\top} \dot{d} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\top} & -R^{\top} d \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} RR^{\top} \dot{R} & RR^{\top} \dot{d} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\top} & -R^{\top} d \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \dot{R} R^{\top} & -\dot{R} R^{\top} d + \dot{d} \\ 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} [\omega_s]_{\times} & v_s \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

Summary

Consider a rigid body moving in space, as shown in the following diagram:

- ω_b is the angular velocity expressed in the body frame, and ω_s is the angular velocity expressed in the fixed frame.
- v_b is the linear velocity of a point at the origin of b and expressed in b .
- v_s is the linear velocity of a point at the origin of s and expressed in s .



Definition: Spatial vs. Body Angular Velocity

Let $R(t)$ be a rotation matrix as seen from some fixed frame, i.e., $R(t) = R_{sb}(t)$, where s is the fixed frame (typically taken to be the world frame) and b is the body frame (attached to the rigid body). Denote $w \in \mathbb{R}^3$ as the angular velocity of the rotating frame.

Spatial angular velocity is fixed-frame vector representation of w (i.e., the angular velocity expressed in the fixed-frame coordinates which is typically defined to be the world frame) and is defined as:

$$[\omega_s]_{\times} = \dot{R}R^{-1} = \dot{R}R^{\top} \in \mathfrak{so}(3)$$

Body angular velocity is body-frame vector representation of w (i.e., the angular velocity expressed in the body frame) and is defined as:

$$[\omega_b]_{\times} = R^{-1}\dot{R} = R^{\top}\dot{R} \in \mathfrak{so}(3)$$

It may be helpful to note that the product $\dot{R}_{sb}R_{sb}^{-1}$ (for spatial angular velocity) is independent of b and the product $R_{sb}^{-1}\dot{R}_{sb}$ (for body angular velocity) is independent of s .

It also might be important to note that ω_b is *not* the angular velocity relative to a moving body frame, but rather it is relative to a *stationary* frame that instantaneously coincides with moving body frame. Otherwise the angular body velocity would always be zero.

Definition: Spatial vs Body Linear Velocity

Spatial linear velocity is the instantaneous velocity of the point (if the rigid body was sufficiently large) located at the fixed-frame origin, expressed in the fixed frame. It is defined as:

$$v_s = \dot{d} - \omega_s \times d = \dot{d} - \dot{R}R^{\top}d$$

Note: v_s is *not* the linear velocity of the body-fixed frame origin expressed in the fixed frame that quantity would simply be $\dot{d}$.

Body linear velocity is the velocity of a rigid body with respect to the (instantaneous) body frame. It is defined as:

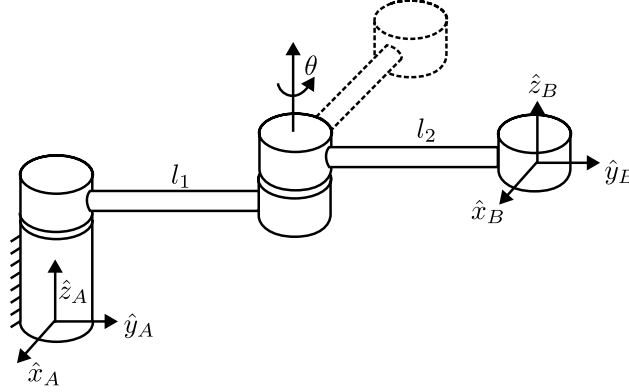
$$v_b = R_{sb}^{-1} \dot{d} = R^\top \dot{d}$$

Derivations for these terms can be obtained through our derivation of the spatial and body twists (note, this was done in previous lectures):

$$\begin{aligned}\hat{\xi}_s &= \dot{g}_{sb} g_{sb}^{-1} \\ \hat{\xi}_b &= g_{sb}^{-1} \dot{g}_{sb}\end{aligned}$$

Example

(Example 2.5 from MLS Chapter 2.4)



We can derive the configuration of this robot as:

$$\begin{aligned}
 g(t) &= g_{at}g_{tb} \quad (\text{Note: } t \text{ represents the temporary frame at the joint location}) \\
 &= \begin{bmatrix} R(\theta(t)) & \begin{bmatrix} 0 \\ l_1 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R(0) & \begin{bmatrix} 0 \\ l_2 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} R(\theta(t))R(0) & \begin{bmatrix} 0 \\ l_1 \end{bmatrix} + R(\theta(t)) \begin{bmatrix} 0 \\ l_2 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} R(\theta(t)) & \begin{bmatrix} 0 \\ l_1 \end{bmatrix} + \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \begin{bmatrix} 0 \\ l_2 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) & -l_2 \sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) & l_1 + l_2 \cos(\theta(t)) \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

The spatial velocity of the rotating rigid body can be solved for as:

$$\xi_s = \begin{Bmatrix} v_s \\ \omega_s \end{Bmatrix} = \begin{Bmatrix} -\dot{R}R^\top p + \dot{p} \\ (\dot{R}R^\top)^\vee \end{Bmatrix}$$

where p denotes the position of the body frame origin in the fixed frame:

$$p(t) = \begin{bmatrix} -l_2 \sin(\theta(t)) \\ l_1 + l_2 \cos(\theta(t)) \end{bmatrix}$$

and R is the orientation of the body frame origin in the fixed frame:

$$R(t) = \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix}$$

If we solve for these expressions you should get:

$$v_s = \begin{Bmatrix} l_1 \dot{\theta} \\ 0 \end{Bmatrix}, \quad \omega_s = \dot{\theta}$$

This computation was derived in a previous class using the homogeneous representation:

$$\begin{aligned} \hat{\xi}_s &= \dot{g}g^{-1} \\ &= \begin{bmatrix} \dot{R}R^\top & -\dot{R}R^\top p + \dot{p} \\ 0 & 0 \end{bmatrix} \quad (\text{Note that the form of this matrix is } \begin{bmatrix} [\omega_s]_\times & v_s \\ 0 & 0 \end{bmatrix}) \\ &= \begin{bmatrix} 0 & -\dot{\theta} & l_1 \dot{\theta} \\ \dot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

We can see how we got this by solving for the terms:

$$\begin{aligned} \dot{R}R^\top &= \begin{bmatrix} -\sin(\theta(t))\dot{\theta} & -\cos(\theta(t))\dot{\theta} \\ \cos(\theta(t))\dot{\theta} & -\sin(\theta(t))\dot{\theta} \end{bmatrix} \begin{bmatrix} \cos(\theta(t)) & \sin(\theta(t)) \\ -\sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} = \begin{bmatrix} 0 & -\dot{\theta} \\ \dot{\theta} & 0 \end{bmatrix} \\ -\dot{R}R^\top p + \dot{p} &= \begin{bmatrix} 0 & \dot{\theta} \\ -\dot{\theta} & 0 \end{bmatrix} \begin{bmatrix} -l_2 \sin(\theta(t)) \\ l_1 + l_2 \cos(\theta(t)) \end{bmatrix} + \begin{bmatrix} -l_2 \cos(\theta(t))\dot{\theta} \\ -l_2 \sin(\theta(t))\dot{\theta} \end{bmatrix} \\ &= \begin{bmatrix} l_1 \dot{\theta} + \cancel{l_2 \cos(\theta(t))\dot{\theta}} - \cancel{l_2 \cos(\theta(t))(\dot{\theta})} \\ \cancel{l_2 \sin(\theta(t))\dot{\theta}} - \cancel{l_2 \sin(\theta(t))\dot{\theta}} \end{bmatrix} = \begin{bmatrix} l_1 \dot{\theta} \\ 0 \end{bmatrix} \end{aligned}$$

Again, note that the interpretation behind spatial velocity is that v_s is the velocity of a point attached to the rigid body as it travels through the origin of the A coordinate frame.

To solve for the body velocity, we follow a similar procedure:

$$\xi_b = \begin{Bmatrix} v_b \\ \omega_b \end{Bmatrix} = \begin{Bmatrix} R^\top \dot{p} \\ (R^\top \dot{R})^\vee \end{Bmatrix}$$

If we solve for these expressions you should get:

$$v_b = \begin{Bmatrix} -l_2 \dot{\theta} \\ 0 \end{Bmatrix}, \quad \omega_s = \dot{\theta}$$

As before, this computation can also be derived using homogeneous representation:

$$\begin{aligned} \hat{\xi}_b &= g^{-1} \dot{g} \\ &= \begin{bmatrix} R^\top \dot{R} & R^\top \dot{p} \\ 0 & 0 \end{bmatrix} \quad (\text{Note that the form of this matrix is } \begin{bmatrix} [\omega_b]_\times & v_b \\ 0 & 0 \end{bmatrix}) \\ &= \begin{bmatrix} 0 & -\dot{\theta} & -l_2 \dot{\theta} \\ \dot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Again, we can check our computations:

$$\begin{aligned} R^\top \dot{R} &= \begin{bmatrix} \cos(\theta(t)) & \sin(\theta(t)) \\ -\sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \begin{bmatrix} -\sin(\theta(t))\dot{\theta} & -\cos(\theta(t))\dot{\theta} \\ \cos(\theta(t))\dot{\theta} & -\sin(\theta(t))\dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & -\dot{\theta} \\ \dot{\theta} & 0 \end{bmatrix} \\ R^\top \dot{p} &= \begin{bmatrix} \cos(\theta(t)) & \sin(\theta(t)) \\ -\sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \begin{bmatrix} -l_2 \cos(\theta(t))\dot{\theta} \\ -l_2 \sin(\theta(t))\dot{\theta} \end{bmatrix} = \begin{bmatrix} -l_2 \dot{\theta} \\ 0 \end{bmatrix} \end{aligned}$$

The body velocity should be interpreted by imagining the velocity of the origin of the B frame, as seen by the B coordinates, so linear velocity is always in the $-x$ direction and angular velocity is always in the z direction (for this problem!).